

(storage) and sends this file to the interpreter. The interpreter then runs the PHP code, fetches the DB data (if required) and puts it into HTML tags. Once the interpreter completes its tasks, it sends the result back to the server, which sends this data to the client (browser) that made a request for this page.

MVC architecture with PHP

The Model-View-Controller concept involved in software development evolved in the late 1980s. It's a software architecture built on the idea that the logic of an application should be separated from its presentation. A system developed on the MVC architecture should allow a front-end developer and a back-end developer to work on the same system without interfering with each other.

Model

Model is the name given to the component that will communicate with the database to manipulate the data. It acts as a bridge between the View component and the Controller component in the overall architecture. It doesn't matter to the Model component what happens to the data when it is passed to the View or Controller components.

The code snippet for running *first_model.php* is:

```
<?php
class Model
{
    public $string;
    public function __construct()
    {
        $this->string = "Let's start php with MVC";
    }
}
```

View

The View requests for data from the Model component and then its final output is determined. View interacts with the user, and then transfers the user's reaction to the Controller component to respond accordingly. An example of this is a link generated by the View component, when a user clicks and an action gets triggered in the Controller.

To run *first_view.php*, type:

```
<?php
class View
{
    private $model;
    private $controller;
    public function __construct($controller,$model)
    {
        $this->controller = $controller;
        $this->model = $model;
```

```
}
public function output()
{
    return "<p>" . $this->model->string . "</p>";
}
}
```

Controller

The Controller's job is to handle data that the user inputs or submits through the forms, and then Model updates this accordingly in the database. The Controller is nothing without the user's interactions, which happen through the View component.

The code snippet for running *first_controller.php* is:

```
<?php
class Controller
{
    private $model;
    public function __construct($model)
    {
        $this->model = $model;
    }
}
```

A simple way to understand how MVC works is given below.

- 1) A user interacts with View.
- 2) The Controller handles the user input, and sends the information to the model.
- 3) Then the Model receives the information and manipulates it (either saving it or updating it by communicating with the database).
- 4) The View checks the state of the Model and responds accordingly (lists updated information).

In the following code snippet, by running *first_example.php* we can see our MVC architecture work with PHP.

To run *first_example.php*, you must type:

```
<?php
$model = new Model();
$controller = new Controller($model);
$view = new View($controller, $model);
echo $view->output();
?>
```



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